

Real-Time AI Fencing Coach: An Edge-Computing Approach to Action Quality Assessment and Instant Feedback

Luther Wei*
weiluther@gmail.com
National Taiwan University
Taipei, Taiwan

You-Chen Lai†
youchen.thomas.lai@gmail.com
National Taiwan University
Taipei, Taiwan

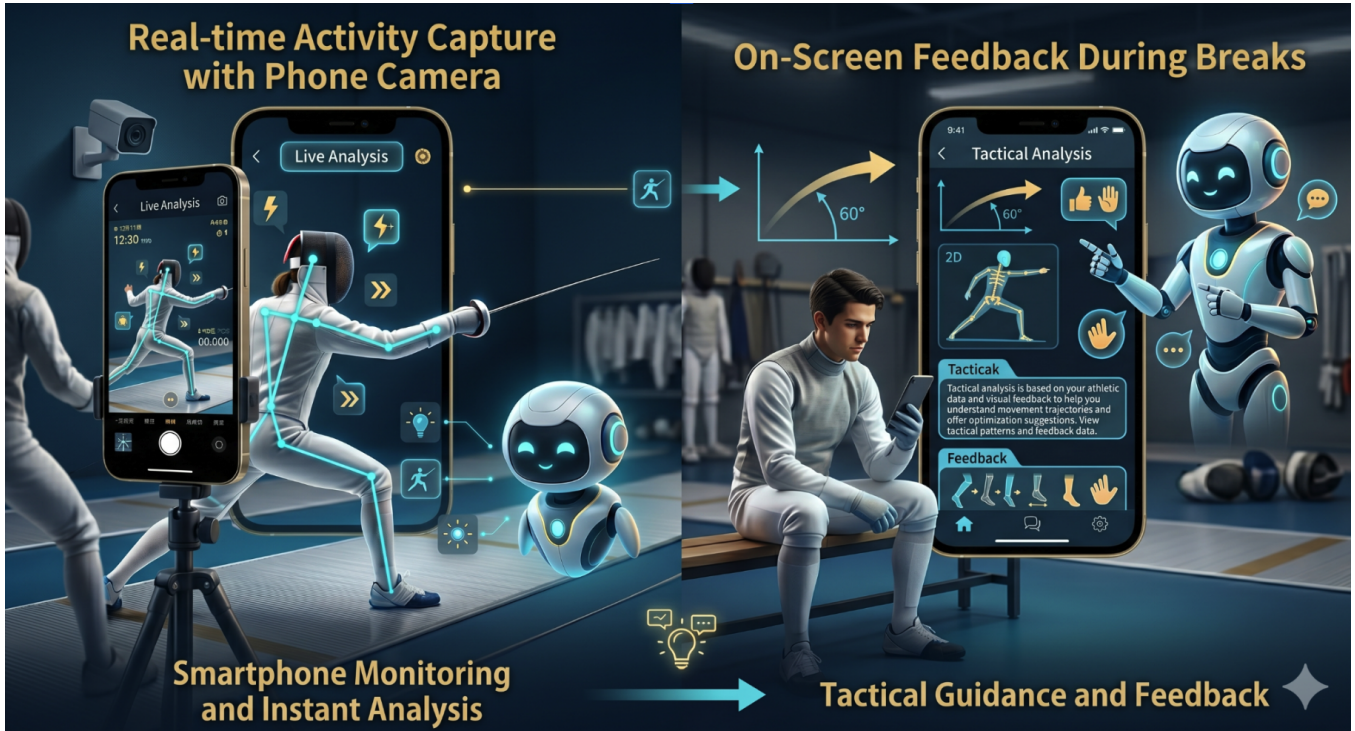


Figure 1: System overview of the AI Fencing Coach pipeline.

Abstract

Fencing requires precise posture, footwork, and timing, yet novice fencers may struggle to recognize technique errors during independent practice. Much existing fencing-computing work emphasizes action recognition or expert analysis, leaving open how fencing-specific feedback should support learners during and after solo practice. We present AI Fencing Coach, a native mobile system that combines on-device pose estimation, FenceNet footwork recognition, and interpretable geometric and temporal heuristics to provide prioritized visual and spoken feedback during practice, followed by detailed post-session review. We conducted a preliminary mixed-methods study of our system. Two participants contributed post-use ratings, and retained records documented unaided self-review, AI-assisted practice, and detailed post-task interviews. Descriptive ratings were highest for error awareness, post-review usefulness, and perceived advantage over self-review ($M=4.75$ for each), and lowest for timing accuracy and correction understandability ($M=3.25$

for each). These findings suggest that combining immediate and post-session feedback can support perceived movement awareness and self-directed practice, while identifying feedback timing and recognition reliability as important design challenges.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; *Interactive systems and tools*.

Keywords

Human-Computer Interaction, Action Quality Assessment, Fencing, Edge AI, Pose Estimation

1 Introduction

Fencing is a highly dynamic combat sport that demands precise coordination, balance, and explosive footwork. A slight deviation in posture—such as dropping the guard during a retreat or overextending the knee during a lunge—can lead to severe tactical disadvantages or physical injury. While one-on-one coaching remains the

*Both authors contributed equally to this research.

†Both authors contributed equally to this research.

gold standard for correcting these micro-movements, it is resource-intensive and often inaccessible for continuous daily practice. Consequently, novice fencers struggle to develop accurate kinesthetic awareness, often internalizing incorrect muscle memory.

To address the scarcity of expert supervision, recent literature has explored computer-vision-based Action Quality Assessment (AQA) systems for sports. However, existing digital fencing coaches and sports analysis tools suffer from several critical limitations that hinder effective human-computer interaction (HCI) during training:

- **The Black-Box Dilemma:** Advanced fencing analysis systems predominantly rely on deep learning architectures. For instance, recent works employ Temporal Convolutional Networks (TCNs) for fine-grained footwork classification, such as the FenceNet architecture [10], or utilize instance segmentation models like YOLACT++ for tracking sword tips [4]. While highly accurate in their respective recognition tasks, these end-to-end models act as opaque black boxes. They cannot explain *why* a movement is wrong or provide actionable, joint-specific corrections (e.g., "your knee angle was 90°"), leading to user frustration and a lack of agency.
- **Hardware Accessibility:** Professional biomechanical evaluation often requires expensive multi-camera setups (e.g., requiring 24 4K cameras [4]) or cumbersome wearable sensors that interfere with real-world matches [9, 10].
- **Scarcity of Real-Time Fencing Coaching:** Despite the rapid, high-speed nature of fencing where immediate error correction is critical, accessible real-time coaching systems remain severely scarce. While broader HCI research highlights the importance of feedback representation in movement learning (e.g., [1–3]) These systems are typically designed for offline review or slower skill acquisition. Current systems fail to apply these principles to the instantaneous demands of combat sports. Consequently, a significant gap remains in translating HCI feedback theories into zero-latency, real-time coaching tools capable of addressing fencing-specific actions, terminology, and error priorities using commodity smartphones.

To overcome these interaction barriers, we present the *AI Fencing Coach*, a mobile-native interactive system designed for real-time, explainable, and accessible training. By leveraging edge computing (Google ML Kit) and a custom geometric heuristic engine, our system acts as a vigilant digital tutor. It tracks 3D skeletal landmarks locally on the user's smartphone and applies domain-expert biomechanical rules to provide zero-latency, actionable voice corrections (e.g., "Raise your guard").

The primary contributions of this paper are:

- (1) A novel edge-AI system architecture for real-time fencing posture assessment.
- (2) An explainable, rule-based heuristic engine capable of translating raw kinematic data into instantaneous, actionable audio feedback.
- (3) A preliminary user study evaluating the impact of explainable, zero-latency feedback on fencers' user experience, usability, and training autonomy.

2 Related Work

2.1 Computer Vision and Footwork Recognition

The application of computer vision in fencing has evolved from sensor-heavy setups to streamlined pose estimation. Malawski and Kwolek pioneered this by introducing the Fencing Footwork Dataset (FFD), which utilized Kinect depth sensors and inertial measurement units (IMUs) to recognize dynamic fencing actions [6]. Building upon the FFD without the need for depth or IMU data, Zhu et al. proposed *FenceNet*, a spatial-temporal architecture leveraging Temporal Convolutional Networks (TCNs) to classify fine-grained footwork techniques using only 2D skeleton data [10]. While these approaches achieve high accuracy in action recognition, they predominantly focus on offline classification rather than providing real-time posture correction for users.

2.2 Markerless Tracking of Fencing Weapons

Tracking the fencing sword presents unique challenges due to its extremely rapid movements and narrow profile, which often result in severe motion blur in standard video recordings. Traditional tracking methods required cumbersome setups, such as applying reflective tape to the sword tip or utilizing massive 24-camera arrays. To address this, Sawahata et al. developed an instance segmentation-based markerless tracking system using normal cameras [8]. Their system employs YOLACT++ to detect the entire sword and incorporates a TCN to predict the tip's location when it eludes detection [8]. Although effective for trajectory visualization, this method focuses on the weapon rather than the fencer's biomechanical posture.

2.3 Sensor-Assisted Training and Virtual Reality

To augment motor learning, researchers have explored systems requiring additional hardware. Niță and Magyar developed an IoT-based training device, evaluated at the ACS Floreta Fencing Club, which requires fencers to wear a back-mounted gyroscope to monitor torso angular velocity [7]. Their system successfully improved balance and movement control by providing real-time visual (LED) and haptic (smartwatch) feedback [7].

In the realm of immersive technology, Dreimane and Zālīte-Supe proposed an instructional design map for Virtual Reality (VR) fencing training to gain tactical experience [5]. However, VR creates a digital environment that is completely closed off from the physical world [5]. Furthermore, the requirement of wearing VR headsets and holding specialized controllers [5] restricts fencers from executing explosive, realistic footwork safely on an actual piste.

In contrast to these approaches, our *AI Fencing Coach* requires zero wearable equipment. Instead of isolating the user in a simulated world or attaching physical sensors to their body, our system augments real-world practice. Using purely markerless mobile pose estimation and auditory real-time feedback, fencers can perform their natural movements on a physical piste without any hardware encumbrance.

3 Formative Study and Analysis

We conducted formative work to identify what an AI fencing coach should say, when it should say it, and which feedback should be

eligible in different practice contexts. This formative study focused on translating coaching knowledge and self-practice breakdowns into design requirements.

3.1 Corpus and Participants

The formative corpus combined coach-facing and fencer-facing evidence. The coach-facing material was an elicitation spreadsheet in which coaches described how they would instruct actions observed in a fencing match video. Each response could include up to four errors, and each error block asked for an error name, the cause or consequence of the problem, the oral cue the coach would say, and a drill or practice activity for improvement.

3.2 Procedure and Analysis

For the coach elicitation, coaches watched fencing actions in a match video and described the errors they observed. We treated each completed error block as an instruction unit containing four components: the observed movement problem, the cause and consequence of the error, the short correction a coach would say, and the practice activity used to improve it. These responses were then normalized into twelve playbook categories. Each category preserved three levels of feedback: a diagnostic explanation for review, a short imperative cue for live practice, and a practice suggestion for follow-up work.

We also checked the implementation as a trace from formative requirements to system behavior. The playbook weights are read by the system and used by the feedback scheduler, which combines coach-derived priority with dynamic factors such as persistence, novelty, aging behind other cues, repeated-speech penalty, and learner focus. Practice modes and user settings further filter which errors can appear.

3.3 Findings

3.3.1 Coach feedback should be layered. The coach-derived material showed that instruction cannot be reduced to a simple label such as "wide step" or "guard dropped." Coaches connect an observed error to a cause, a short correction, and a practice activity. For example, overly forceful thrusting may be explained as excessive shoulder effort, cued as relaxing the shoulder or using the fingers, and paired with a drill for blade control. This structure motivated the playbook format: each error stores a diagnosis, short cue, practice suggestion, and priority weight.

This distinction shaped the system design. Live feedback should remain short enough to use during movement, while post-session review should preserve the reasoning behind the cue. For example, "Stance Too High" is spoken as "Stay lower," but the review explains how a high stance reduces balance and readiness to change direction. Similarly, "Foot Before Hand" is cued as "Hand first," while the diagnosis explains why moving the foot before creating weapon threat makes an attack easier to read.

3.3.2 Feedback should be queued by coaching priority and practice state. Several errors can occur in the same movement window, but learners cannot act on all of them at once. The coach-derived playbook therefore assigns weights as an initial approximation of pedagogical priority. Foundational form issues, such as stance

height, guard position, lunge overextension, incomplete arm extension, and narrow steps, receive higher weights than less central or more context-dependent errors.

The scheduler turns these weights into a dynamic cue queue rather than a fixed severity list. It boosts persistent errors, gives novelty to errors that have not yet been spoken, ages skipped issues so they do not disappear, penalizes repeated speech, and raises learner-selected focus categories. This treats priority as contextual: coach weight starts the ranking, but the current practice state affects what is most useful to say next.

3.3.3 Practice mode should scope feedback. The formative evidence also showed that different practice contexts require different feedback scopes. Footwork practice should emphasize stance, bounce, step width, and center-of-mass control. Target practice should emphasize attack coordination, hand-before-foot timing, arm extension, lunge control, and guard position. Free bouting requires broader feedback but must avoid overloading the learner during mixed tactical movement.

The current system implements this logic through three modes: Footwork, Target Practice, and Free Bouting. Some errors, such as `foot_before_hand` and `incomplete_arm_extension`, are restricted to Target Practice, while posture, guard, balance, step, and parry-related errors remain more broadly available. The settings screen also allows learners to emphasize, mute, or restrict feedback categories.

3.4 Design Requirements

The formative study led to five design requirements. First, the system should represent coach instruction as layered feedback, separating short live cues from diagnostic explanation and practice suggestions. Second, it should reduce the diagnostic burden of solo practice by helping learners notice and name errors they may not identify from video alone. Third, it should queue feedback by coaching priority and current practice state rather than simply reporting every detected error. Fourth, it should scope feedback by practice mode so that learners receive cues relevant to the drill they are performing. Fifth, it should preserve learner contestability by linking feedback to reviewable evidence and allowing learners to pause, mute, focus, or question system output.

These requirements position the formative study as a translation from coaching practice to interaction design. The purpose is not to prove learning gains, but to define what the AI fencing coach should preserve from human instruction, which self-practice pain points it should address, and where the current prototype still requires refinement.

4 System Design

AI Fencing Coach is designed to reorganize solo video practice into a continuous notice-interpret-act-verify loop. In ordinary self-review, the learner must remember a movement, inspect the recording, decide what is wrong, and translate that judgment into a correction. The prototype distributes this work across three feedback layers: a live camera view makes the analyzed body and action visible, short visual and spoken cues support the next attempt, and a persistent review record supports slower interpretation after practice.

Its core interaction mechanism is therefore not action classification alone, but the conversion of a continuous practice stream into sparse correction during movement and inspectable evidence after movement.

Based on the domain challenges and HCI literature identified in previous sections, we derived five primary design goals (DG) to guide the system's development. These goals aim to reduce diagnostic work without treating the AI as the final, unquestionable authority.

4.1 Design Goals

DG1: Add diagnostic support without replacing self-assessment. The system should help learners notice and name movement features while still letting them compare automated output with what they felt and saw. During practice, it shows the camera image, optional skeleton overlay, current action state, and detected issues rather than only a hidden score. After practice, recognized-action counts and a cue timeline allow learners to compare feedback with memory and other evidence. This supports diverse learning stages: novices may need explicit labels and priorities, while experienced fencers might use feedback as confirmation or a point of disagreement.

DG2: Separate immediate action from later explanation. Live feedback should answer only what to change on the next attempt. The system uses short directional cues such as "Stay lower" or "Shorten the step," speaks at most one correction at a time, and displays no more than three issues. Broader diagnosis, recurrence, and practice suggestions are moved to the review layer. This reflects the cognitive load during practice: actionable cues (e.g., step-size) aid immediate adjustment, while complex form errors (e.g., defensive actions) require deeper post-hoc explanation. Thus, low interruption and full understanding are treated as separate outcomes.

DG3: Preserve the temporal provenance of feedback. A correction is useful only when the learner can identify the movement episode it refers to. The system records each accepted cue with a frame index, converts it into relative session time, and exposes recent and post-session timelines. Selected-video mode also preserves frame-level analysis states for optional annotation.

DG4: Make uncertainty and failure visible enough to question. The interface distinguishes model loading, target search, stance checking, active analysis, pause, and review states. When a non-idle action is recognized, the action label includes classifier confidence and processing rate. Internal diagnostics also record pose, classifier, total processing time, and estimated dropped frames. These states make the recognition process more visible than a single authoritative diagnosis.

DG5: Preserve learner and coach authority. Learners can pause analysis, silence speech, reset or resume sessions, emphasize or mute error categories, restrict feedback categories, decide whether to use a generated summary, and delete stored sessions. A deterministic fencing playbook remains available when no language-model service is selected or a request fails. These controls make automation configurable rather than compulsory. The system remains a

practice aid, not a coach replacement: it has no coach-facing validation workflow, and its thresholds should not be treated as universal definitions of correct technique.

4.2 Interaction Flow

The system provides two entry points into the same feedback model: **Realtime** for practice feedback and **Postgame** for analysis of selected videos. A separate **History** area supports comparison across saved sessions, while **User Settings** controls practice mode, target position, pose engine, language, speech, feedback focus, skeleton overlay, summaries, and video export. This separation lets learners set attention and privacy preferences before entering the camera-based interaction.

In Realtime mode, the learner confirms the configuration and starts the back camera. The system searches for one intended fencer; if two people are visible, the configured left or right position selects the target while the other person may remain as opponent context. The interaction moves through IDLE, CHECKING, and ACTIVE states. IDLE searches for a fencing stance, CHECKING waits for a short en garde sequence, and ACTIVE sends movement frames into the action and biomechanical analysis pipeline. Standing upright, turning away, or losing the target returns the system to IDLE, while short pose dropouts are bridged to preserve context.

During active practice, the upper display shows the camera and optional skeleton overlay. An action indicator reports the recognized action and confidence, while the lower panel presents one primary issue and up to two secondary cues. If speech is enabled, Android text-to-speech releases one eligible cue at a time, subject to cooldowns. The learner can continue practicing without acknowledging each cue, pause analysis, toggle speech, reset state, or finish the session.

Finishing practice opens a review report summarizing elapsed and active time, model checks, retained cues, most frequent action, action counts, recurring issues, and a relative cue timeline. Each issue may include a short cue, diagnosis, and suggested drill. Reports are saved on device and can later be filtered, opened, summarized, selected for recap, or deleted.

Postgame mode applies the same target, action, heuristic, scheduling, and report-building logic to uploaded videos. It shows processing progress, then produces a report and can optionally export an annotated video. Together, Realtime and Postgame support complementary use: live cues guide the next repetition, while review supports inspection and explanation.

Across both modes, the system is iterative rather than terminal. It does not assign a pass/fail grade, but offers bounded observations that learners can act on, revisit, compare, or reject according to drill context, skill level, weapon, and coach intent.

4.3 Feedback Scheduling and Prioritization Algorithm

The formative study highlighted that rapid auditory feedback is ineffective if learners cannot cognitively map the cue to the referenced action. Therefore, our system treats feedback delivery as a dynamic scheduling problem, coordinating among error detection, priority ranking, and attention budgeting.

Let $E_{\text{det}}(t)$ represent the set of biomechanical issues detected within a short temporal window at time t . The scheduling algorithm processes this set through three sequential stages: Filtering, Dynamic Scoring, and Cooldown Gating.

1. *Filtering and Candidate Selection.* The system first filters $E_{\text{det}}(t)$ based on the user's active practice mode and muted categories. The remaining issues form the candidate set $E_{\text{cand}}(t)$.

2. *Dynamic Priority Scoring Function.* Rather than relying on a static severity list, the scheduler computes a dynamic priority score $S(e, t)$ for each candidate issue $e \in E_{\text{cand}}(t)$. The scoring function balances baseline severity, temporal aging (to prevent starvation), novelty, and learner emphasis, while penalizing repetitive cues:

$$S(e, t) = \alpha \cdot \omega_{\text{base}}(e) + \beta \cdot P(e, t) + \gamma \cdot T_{\text{wait}}(e, t) + \delta \cdot \Omega_{\text{user}}(e) - \lambda \cdot R(e, t)$$

where:

- $\omega_{\text{base}}(e)$: Base severity weight defined in the fencing playbook.
- $P(e, t)$: Persistence of the error over recent frames.
- $T_{\text{wait}}(e, t)$: Time elapsed since the error was detected (aging factor).
- $\Omega_{\text{user}}(e)$: Binary flag indicating if the learner explicitly emphasized this category.
- $R(e, t)$: Repetition penalty to suppress redundant feedback.

This dynamic state ensures that persistent or learner-emphasized issues rise in priority, while repeated cues are penalized to maintain engagement.

3. *Attention Budgets and Cooldown Gating.* To manage the learner's cognitive load, the visual and spoken channels operate under different attention budgets and strict temporal constraints. The visual interface renders the top $k \leq 3$ issues ranked by $S(e, t)$.

For the auditory channel, the system serializes output to a single cue e^* (where $e^* = \arg \max S(e, t)$) subject to three temporal gates:

- (1) **Global Delay:** The time since the last spoken cue must exceed a global threshold ($\Delta t_{\text{global}} \geq 1.2\text{s}$) to prevent overlapping audio.
- (2) **Same-Error Cooldown:** The time since error e^* was last announced must exceed a specific cooldown ($\Delta t_{e^*} \geq 4.0\text{s}$).
- (3) **Pending Window:** The issue must have been detected within a recent validity window ($T_{\text{wait}}(e^*) \leq 5.0\text{s}$). If a cue expires before clearing the cooldowns, it is dropped from the live queue to prevent temporal ambiguity (i.e., feedback arriving after the triggering movement has passed).

These constants are current implementation choices for the prototype, and future work will empirically validate capture-to-cue latency and test whether learners can reliably identify the referenced repetition under different delivery strategies (e.g., immediate vs. movement-boundary).

4. *Live Abstraction and Retrospective Traceability.* To accommodate the live auditory bottleneck, the system abstracts the winning cue e^* into a short, directional phrase (e.g., "Stay lower"). The full diagnostic context—including the raw action count, joint deviations, and suggested drills—is archived for review. This preserves a traceable path from the instantaneous live cue to the comprehensive

post-session explanation, allowing learners to inspect skipped or ambiguous feedback without disrupting their active practice flow.

5 Implementation

5.1 Core Algorithmic Engine: Biomechanical Heuristics

Instead of relying on a black-box end-to-end classifier, our system evaluates fencing posture through a deterministic, explainable heuristic engine. The algorithm continuously processes a sliding temporal window of skeletal frames to compute real-time kinematic parameters, such as joint angles, relative limb displacements, and body-scaled proportions.

Let $S_t = \{p_1, p_2, \dots, p_k\}$ denote the set of 3D joint coordinates extracted at frame t , where $p_i = (x_i, y_i, z_i)$. To evaluate the biomechanical correctness of a movement, the algorithm maintains a temporal sliding window $W = [t - \Delta t, t]$ of recent skeletons. We define two primary types of algorithmic triggers: **Spatial-Geometric Triggers** and **Temporal-Persistent Triggers**.

1. *Spatial-Geometric Triggers (Instantaneous State).* These triggers evaluate the kinematic constraints within a single frame t . The system dynamically computes the angle θ between three adjacent joints (e.g., hip, knee, and ankle) using the dot product of their vectors. For instance, the *Lunge Overextension* error is triggered if the front knee angle drops below a safe biomechanical threshold:

$$\theta_{\text{knee}} = \arccos \left(\frac{\vec{v}_{\text{thigh}} \cdot \vec{v}_{\text{calf}}}{|\vec{v}_{\text{thigh}}| |\vec{v}_{\text{calf}}|} \right) < 120^\circ$$

Similarly, step width is evaluated by normalizing the distance between the front and rear ankles (d_{ankles}) against a dynamic body-scale proxy (d_{shoulder}) to adapt to different user heights. If $d_{\text{ankles}} > 3 \times d_{\text{shoulder}}$, a *Step Too Wide* error is flagged.

2. *Temporal-Persistent Triggers (Time-Series State).* Fencing actions involve dynamic transitions; thus, certain postures (e.g., dropping the guard during a retreat) are only considered errors if they persist over time. The algorithm evaluates the continuity of an erroneous state across the sliding window W . For the *Dropped Guard* error, the algorithm checks the relative vertical position of the weapon-holding wrist against the pelvis:

$$\sum_{i \in W} \mathbb{I}(y_{\text{wrist}}^{(i)} < y_{\text{pelvis}}^{(i)}) > \tau_{\text{frames}}$$

where $\mathbb{I}$ is the indicator function and τ_{frames} is the temporal threshold (e.g., 5 frames). This temporal gating prevents false positives caused by natural, momentary arm swings.

Table 1 summarizes the 12 core deterministic rules implemented in the engine. By structuring the algorithm with explicit geometric and temporal constraints, the system ensures that every triggered error is directly tied to a specific kinematic anomaly, enabling the generation of highly explainable, joint-specific feedback.

5.2 Visual, Spoken, and Post-Session Feedback

The feedback scheduler converts simultaneous rule detections into a bounded set of outputs. For each active error, it computes a priority score from a playbook-defined base weight, an aging bonus of 2 per skipped opportunity, a persistence bonus of 0.25 for up to

Table 1: Implemented biomechanical feedback rules and their algorithmic operating conditions.

Feedback Rule	Algorithmic Trigger Condition	Temporal & Contextual Gating
Excessive bounce	$\Delta y_{\text{pelvis}} > 0.33 \times \text{body_height}$	Evaluated over $W = 5$ frames after a 15-cycle warm-up
Lunge overextension	$\theta_{\text{front_knee}} < 120^\circ$	Instantaneous evaluation on active frames
Dropped guard	$y_{\text{weapon_wrist}} < y_{\text{pelvis}}$	Persistent over $\tau = 5$ frames (or 10 in free bout)
Foot before hand	$\Delta x_{\text{front_ankle}}$ peaks before Δx_{wrist} ; both $> 0.01 \times \text{scale}$	Requires target-practice mode + offensive action state
Stance too high	Mean $\theta_{\text{front_knee}} > 160^\circ$	Averaged over $W = 3$ frames
Incomplete arm ext.	$\theta_{\text{weapon_elbow}} < 155^\circ$	Requires target-practice mode + offensive action state
Over-parrying	$\Delta x_{\text{wrist_sweep}} > 3 \times d_{\text{shoulder}}$	Evaluated over $W \geq 5$ frames
Step too wide	$d_{\text{ankles}} > 3 \times d_{\text{shoulder}}$	Instantaneous evaluation
Step too narrow	$d_{\text{ankles}} < 1.2 \times d_{\text{shoulder}}$	Instantaneous evaluation
CoM too far forward	$\theta_{\text{spine_tilt}} > 15^\circ \vee \theta_{\text{shoulder_tilt}} > 15^\circ$	Instantaneous geometric proxy for Center of Mass (CoM)
CoM too far backward	$\theta_{\text{spine_tilt}} < -10^\circ \vee \theta_{\text{shoulder_tilt}} < -15^\circ$	Instantaneous geometric proxy for Center of Mass (CoM)
Hand too high	Angle of wrist above elbow $> 60^\circ$	Instantaneous evaluation

eight active cycles, a novelty bonus of 0.75, a focus-category bonus of 4, and a repetition penalty of 1 for up to three prior spoken instances. The scheduler removes muted or mode-incompatible errors, displays at most three visual cues, and selects at most one cue for speech. It enforces a 1.2-second global speech interval and a 4-second per-error cooldown. An issue remains pending for up to 5 seconds after its most recent detection, limiting how long stale advice remains eligible for speech.

Labels, short corrective cues, diagnoses, drills, and base priorities are loaded from a bilingual JSON playbook. During practice, the interface presents the recognized action, confidence, processing state, and prioritized correction cards. Android text-to-speech uses flush semantics so that a newly selected cue replaces queued speech. Users can pause analysis, disable voice, focus on one error category, mute categories, or terminate the session. When no visual rule is active the interface shows "Good Technique"; this means only that no implemented rule fired on the current evidence, not that the movement was biomechanically correct.

The application logs a spoken cue, or otherwise the first visual cue, while deduplicating the same label within a classifier stride. The live history keeps five recent cues, the session timeline keeps up to 60, and the report displays the 20 most recent entries. Room stores session metadata, aggregate classification-window counts, and cue history. The post-session report combines these records with playbook explanations and drill suggestions. If enabled, annotated-video export renders the sampled analysis over the original recording through Media3. Several failure paths are handled explicitly. Missing pose or FenceNet assets produce a model-unavailable state, optional language-model failure falls back to the deterministic report, and short target loss is bridged by the tracker. However, exceptions in the live camera analyzer are currently suppressed to protect the session from a crash and are not surfaced to the learner. Better diagnostic logging and a visible degraded-mode indicator are needed before the system can support reliability claims.

6 User Experience Study

To evaluate the human-computer interaction aspects of the AI Fencing Coach, we conducted a preliminary summative user experience

study. The study focused on how learners use AI feedback compared to unaided self-review, their comprehension and ability to act on the feedback, and their trust in the system when encountering varying temporal accuracy and recognition quality.

6.1 Research Questions

The study investigated the following research questions:

- (1) **RQ1:** What differences emerge between the errors fencers identify through unaided video self-review and those they report after AI-assisted feedback?
- (2) **RQ2:** Which properties of live feedback make a correction actionable without unnecessarily interrupting practice?
- (3) **RQ3:** How do cue timing, action-recognition errors, and post-session evidence shape appropriate reliance on AI-generated fencing feedback?

6.2 Participants and Procedure

Participants consisted of two fencing novices who interacted with the AI Fencing Coach. Detailed interview records were retained and analyzed for two participants (P1 and P2). They were asked to perform standard fencing footwork drills, including simple step back, step forward, lunge, and step lunge.

The study procedure consisted of four phases:

- (1) **Unaided Self-Review:** Participants performed a short fencing movement set without feedback and then reviewed their recorded video to identify problems.
- (2) **AI-Assisted Practice:** Participants repeated the movements while receiving real-time spoken cues through earphones, such as "stance too high" and "step too wide."
- (3) **Feedback-Guided Adjustment:** After receiving adjustments from the real-time AI system, participants performed the same drills again and compared their posture with the pre-session recording.
- (4) **Post-Session Analysis:** Participants reviewed the system's post-session video report, which provided timestamped issues and summaries. This was followed by a semi-structured interview regarding their user experience, trust, and comprehension.

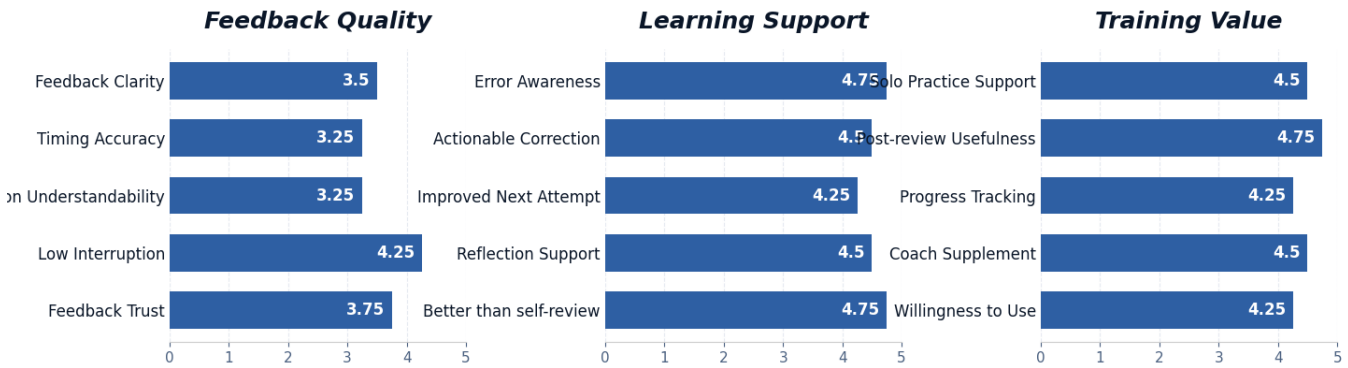


Figure 2: Main results

Post-session, participants also completed a Likert-scale questionnaire (1–5) assessing usability, user experience (UX), and usefulness.

6.3 Results

6.3.1 Questionnaire Feedback. In Figure 2, initial feedback showed that Error Awareness, Post-Review Usefulness, and Better than Self-Review scored the highest, both above 4.75/5.0. Users particularly appreciated the explainability of the feedback, feeling more “in control” compared to traditional video playback tools.

6.3.2 Error Awareness and Diagnostic Support (RQ1). The unaided self-review phase revealed differing baseline diagnostic abilities among users. P1 independently identified gross errors, such as vertical center-of-mass movement and narrow steps, and could state a corresponding adjustment. In contrast, P2 struggled to name specific problems from the video alone, expressing uncertainty until prompted by the interviewer.

After interacting with the AI Fencing Coach, both participants demonstrated increased error awareness and found the post-session review highly useful. For P2, the system provided an initial diagnostic vocabulary, directly pointing out center-of-mass and step-size issues that they could not confidently see. As P2 noted, the system analysis was more helpful than self-review because it directly pointed out specific problems rather than relying on their own uncertain judgment. For P1, the system offered confirmation of known issues and added finer details, such as identifying unnecessary arm movements as defensive actions. This indicates that AI support does not merely reveal invisible errors; it structures the review process, providing foundational labels for novices while confirming or detailing issues for more self-aware learners.

6.3.3 Actionability and Interruption (RQ2). Both participants successfully acted on concise directional cues during practice. P1 and P2 reported that feedback regarding step size and center of gravity was highly actionable because it implied a clear physical adjustment for the next repetition, such as stepping closer or lowering the stance. Notably, neither participant felt that the earphone audio was a major interruption to their practice rhythm.

However, more complex or unfamiliar feedback, such as corrections related to the weapon arm or defensive actions, was difficult to process in real time. P1 noted that the post-session video analysis was necessary to provide the contextual detail that short spoken cues lacked. The video review added a crucial layer of explanation that helped participants understand the nuances of their movements. This highlights the importance of layering feedback: providing brief, actionable cues during movement, supported by detailed, timestamped explanations during rest or review.

6.3.4 Cue Timing, Recognition Errors, and Trust (RQ3). While participants could act on the feedback, they experienced challenges related to temporal grounding. Both P1 and P2 reported instances where they were unsure whether a spoken cue referred to the movement they were currently executing or a previous repetition. P2 specifically noted that some cues arrived with a delay, making it ambiguous which action the system was correcting. This ambiguity indicates that temporal accuracy is critical for live feedback; if the timing is off, the user cannot confidently link the cue to the physical feeling of the movement.

Furthermore, participants critically evaluated the system’s accuracy rather than trusting it unconditionally. Both P1 and P2 encountered cases where the system misclassified normal hand movements as a “defensive action.” While P1 reinterpreted the diagnosis as somewhat plausible after discussion, P2 explicitly rejected the system’s assessment, maintaining that they had not performed a defensive motion. This demonstrates that an incorrect action classification undermines the associated coaching advice. It emphasizes the need for the system to clearly link feedback to inspectable video evidence, allowing learners to contest, verify, and appropriately rely on the AI’s judgment rather than blindly accepting false positives.

7 Discussion, Limitations, and Future Work

7.1 AI Feedback as Diagnostic Support

Our user study suggests that AI feedback adds useful diagnostic structure to solo fencing practice. Participants valued the system for making errors easier to notice, name, and revisit. Both interviewees acted on concise cues about step size or balance, yet these cues were useful only when they could connect the advice to a specific movement and attempt a correction.

The prototype is therefore best understood as supporting a temporally grounded feedback loop rather than replacing technical judgment. AI feedback helped narrow continuous video into candidate problems, gave learners fencing-specific labels, and preserved recurring issues for later review. AI support should therefore complement self-review by adding evidence, prioritization, and contestable prompts, rather than assuming that automated feedback is always superior.

7.2 Designing Useful Live and Review Feedback

The findings show that actionability depends on compression, while reflection depends on explanation. Short live cues were useful because they reduced the next decision to one possible correction. At the same time, more technical issues were difficult to resolve through audio alone, which helps explain why brief cues could feel actionable while still being insufficiently understandable for later analysis.

This suggests that live and post-session feedback should serve different roles. Live feedback should be timely, limited, and tied clearly to the movement episode that produced it. Review feedback should restore the evidence omitted from the live cue, including the relevant video segment, action label, repeated patterns, and rationale for correction. Future systems should evaluate capture-to-cue latency, mark the movement being referenced, and defer feedback when its referent is no longer clear.

7.3 Contestability and Deployment Boundaries

Appropriate reliance depended less on general trust in AI than on whether a specific cue could be reconciled with memory and video evidence. In our study, P2 explicitly rejected a defensive-action label because no defensive action had occurred. This reveals a key risk: when action recognition is wrong, even well-written coaching advice becomes irrelevant or misleading. A coaching system must therefore expose uncertainty at the level of the correction, preserve the source episode, and allow users or coaches to dismiss or annotate false feedback.

AI Fencing Coach is best positioned between unaided self-review and expert coaching. It can provide consistent attention to a bounded set of visible features and remember repeated cues. However, human coaches remain necessary for deciding which errors matter, whether corrections are technically appropriate, and how tactics or opponent interaction should affect instruction.

7.4 Limitations and Future Work

Study Limitations: The current evaluation was a preliminary, supervised study with a small sample size. The study sequence was fixed (self-review always preceded AI-assisted practice), which means familiarity with the movement or fatigue could be confounding variables. Most importantly, the study measured perceived usefulness and UX, but did not measure objective, coach-blinded technique improvement or long-term retention.

System Limitations: The heuristic layer checks twelve implemented error categories using fixed geometric and temporal thresholds. We have not yet reported a representative benchmark of pose quality, action classification accuracy, or heuristic precision across diverse hardware and lighting conditions. Furthermore, while the

scheduler mitigates cognitive overload, the temporal ambiguity of delayed cues remains a challenge.

Future Work: Future research should first technically benchmark the feedback engine against expert annotations to evaluate false positives, latency, and tracking failures. Subsequent user studies should employ a within-subjects counterbalanced design, comparing AI-assisted training against baseline video review over a longitudinal period. Incorporating direct input from professional coaches to refine the rule thresholds and exploring adaptive feedback timing (e.g., delivering cues strictly at movement boundaries) will be critical next steps.

8 Conclusion

We presented AI Fencing Coach, a native mobile system for supporting fencers when continuous coaching is unavailable. The system operationalizes a layered feedback interaction: an on-device pipeline converts camera-based pose, target tracking, action recognition, and fencing-specific heuristics into one prioritized spoken cue and a small set of visual issues, while timestamped post-practice reports preserve evidence for reflection. In a preliminary mixed-methods study of an early prototype, two participants contributed post-use ratings and detailed retained interviews. Error Awareness, Post-Review Usefulness, and Better than Self-Review had the highest aggregate means; the interviews indicated that brief step-size and balance cues could guide reported next-attempt adjustments. At the same time, delayed cues and action misclassification made otherwise understandable advice difficult to connect to the movement that produced it. Together, these findings identify temporal grounding, layered explanation, prioritization, and contestability as requirements for interpretable AI movement feedback that supports rather than replaces learner and coach judgment. Given the small, partially retained corpus and absence of objective skill measures, this work contributes feasibility evidence and design requirements rather than proof of durable fencing improvement.

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